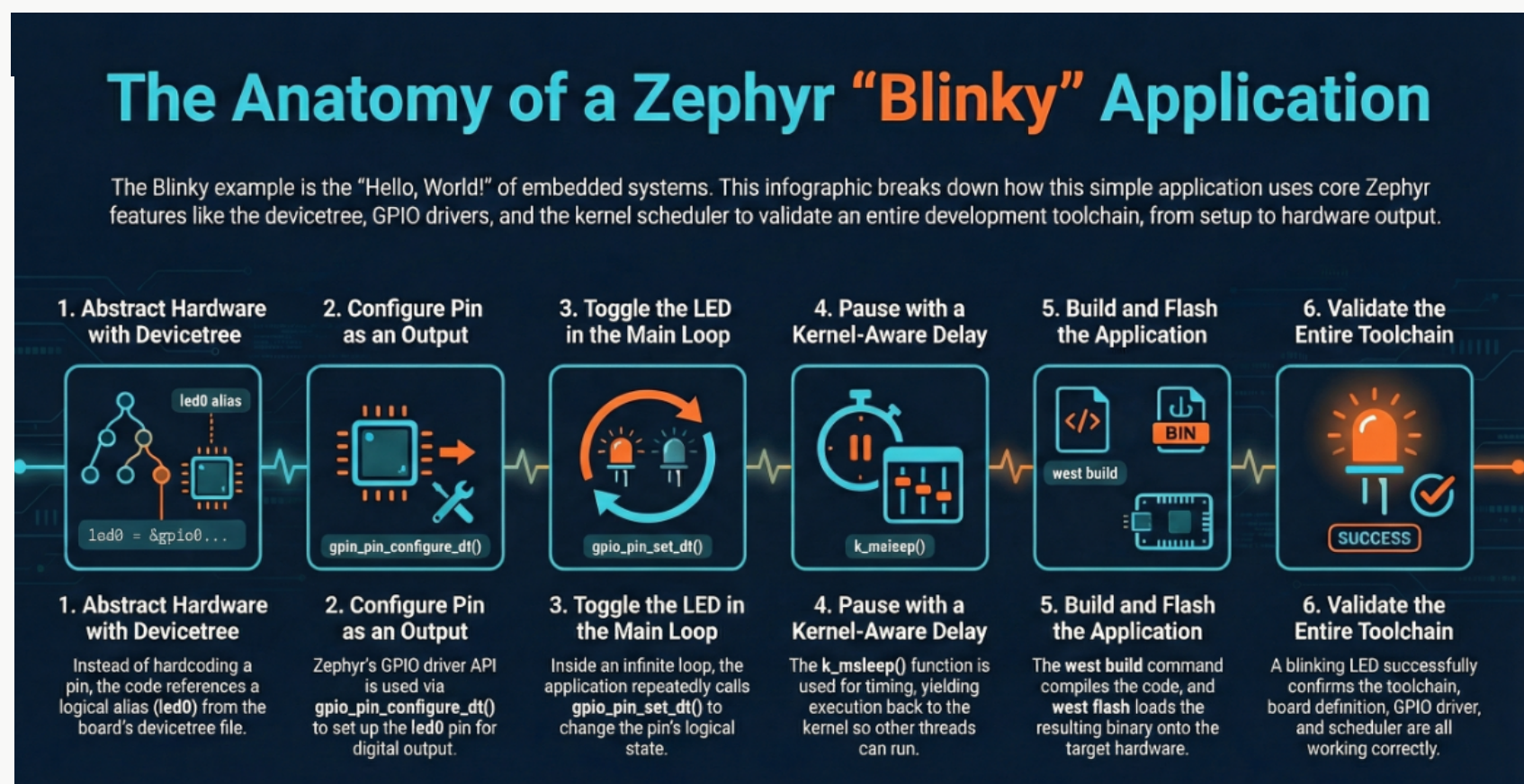


Blinky Example

- GPIO output to drive an LED connected to the board.
- Basic timing with `k_msleep`.
- Project structure and key files.



```

1  zephyr-esp32s3-course > apps > blinky > src > C main.c > main(void)
2
3  #include <stdio.h>
4  #include <zephyr/kernel.h>
5  #include <zephyr/drivers/gpio.h>
6
7  /* 1000 msec = 1 sec */
8  #define SLEEP_TIME_MS 2000
9
10 /* The devicetree node identifier for the "led0" alias. */
11 #define LED0_NODE DT_ALIAS(led0)
12
13 /*
14  * A build error on this line means your board is unsupported.
15  * See the sample documentation for information on how to fix this.
16  */
17 static const struct gpio_dt_spec led = GPIO_DT_SPEC_GET(LED0_NODE, gpio);
18
19 int main(void)
20 {
21     int ret;
22     bool led_state = true;
23
24     if (!gpio_is_ready_dt(&led))
25     {
26         return 0;
27     }
28
29     ret = gpio_pin_configure_dt(&led, GPIO_OUTPUT_ACTIVE);
30     if (ret < 0)
31     {
32         return 0;
33     }
34
35     while (1)
36     {
37         ret = gpio_pin_toggle_dt(&led);
38         if (ret < 0)
39         {
40             return 0;
41         }
42
43         led_state = !led_state;
44         printf("LED state: %s\n", led_state ? "ON" : "OFF");
45         k_msleep(SLEEP_TIME_MS);
46     }
47     return 0;
48 }
49
50
51
52
53

```

Key idea: Blinky is the simplest way to validate your toolchain and hardware.